

2.2 Electrons, bonding and structure

This section introduces the concept of atomic orbitals and develops a deeper understanding of electron configurations linked to the periodic table.

The central role of electrons in ionic and covalent bonding is then studied. The important role of

molecules is studied, including an explanation of polarity and intermolecular forces. Finally, this section looks at how bonding and structure contribute to properties of substances.

2.2.1 Electron structure

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Energy levels, shells, sub-shells, atomic orbitals, electron configuration	
(a) the number of electrons that can fill the first four shells	
(b) atomic orbitals, including: (i) as a region around the nucleus that can hold up to two electrons, with opposite spins (ii) the shapes of s- and p-orbitals (iii) the number of orbitals making up s-, p- and d-sub-shells, and the number of electrons that can fill s-, p- and d-sub-shells	HSW1,7 Development of models to explain electron structure.
(c) filling of orbitals: (i) for the first three shells and the 4s and 4p orbitals in order of increasing energy (ii) for orbitals with the same energy, occupation singly before pairing	Learners are expected to be familiar with the 'electrons in box' representations. HSW1 Development of refined models for electron structure.
(d) deduction of the electron configurations of: (i) atoms, given the atomic number, up to $Z = 36$ (ii) ions, given the atomic number and ionic charge, limited to s- and p-blocks up to $Z = 36$.	Learners should use sub-shell notation, i.e. for oxygen: $1s^2 2s^2 2p^4$.